

Cause-for-Concern: A Domain-Specific Language for Disease Diagnosis and Therapy by Coherence Localization and Gap-Closure

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Abstract

We specify *Cause-for-Concern* (**cfc**; source extension **.cfc**), a domain-specific language for clinical reasoning whose semantics dispenses with the two classical primitives of medicine—the patient’s narrative report and the named disease drawn from a catalogue—and replaces them with a single primitive: the *characterised perturbation* of a patient’s physiological state, whose fate is computed against a forward model and read back as a measurement. The language is self-contained and rests on five facts we prove rather than assume. **(i)** A finite observer of a physiological system cannot resolve its state to a point; a strictly positive resolution floor $\beta > 0$ forces every determined state to be a *cell* of diameter $\geq \beta$, so a diagnosis is the identification of a cell, never a label (*Floor Theorem*). **(ii)** Health is not a distinguished catalogue entry but membership of a *coherent set*, the states in which the system’s oscillator ensemble is internally phase-consistent, quantified by a scalar coherence $\eta \in [0, 1]$ and a disease vector $\mathbf{D} \in [0, 1]^d$; diagnosis is the *localization* of the patient to a cell, provably sufficient exactly when the probe-to-reading map is injective down to the floor (*Localization Theorem*). **(iii)** Because each physiological response is itself a substrate for the next, a probe can be made to *cascade*, so its terminal readout *certifies* the trajectory that produced it: a false account of the state cannot forge it, as it would have to produce products that never formed (*Certificate Theorem*), and the correct halting rule is a *rigidity* condition, not a magnitude threshold. **(iv)** Therapy is closure of the measured *gap* to the nearest coherent state by the sparsest combination of characterised interventions selected *by effect* and not by indication, solved as an ℓ_1 program and executed as a contraction whose fixed point is coherence, with no disease ever named (*Gap-Closure* and *Convergence Theorems*). **(v)** Diagnosis and therapy are one operation, since every characterised perturbation simultaneously informs (its fate is a measurement) and displaces (its action is an intervention); the loop halts precisely when the state is both coherent and rigidly known (*Wellness Theorem*). We give **cfc**’s complete lexical grammar, context-free grammar in EBNF, an *accountability* type system in which no value of zero residue is well-typed and no **close** converges without a localized cell, a small-step operational semantics in which evaluation is measurement—mutating a monotone committed-measurement clock rather than simulating—and proofs of type soundness (progress and preservation), determinism, and scheduler soundness (a closed cell is never re-probed and a convergent therapy is never starved). We specify complexity, a constructive validation protocol requiring no clinical data, and a reference execution model. The result is a deterministic, disease-free, report-free calculus in which a patient’s state is read from a self-certifying physiological program, the therapeutic target is a measured gap, and clinical judgment is relocated from precedent-retrieval to the specification of *intent* over a derived, explorable state.

Keywords: domain-specific language; disease diagnosis; therapy; resolution floor; coherence;

state localization; trajectory certificate; gap-closure; sparse intervention; contraction to a fixed point; accountability type system; operational semantics; measurement-as-evaluation.

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1 Introduction

1.1 Two primitives medicine cannot justify

Clinical practice is organised around two primitives that its own methodology cannot ground. The first is the *patient's report*: a narrative of symptoms, taken as the entry point to diagnosis. The second is the *named disease*: a label drawn from a finite nosological catalogue, against which the report is matched. Both are load-bearing, and both are fragile. A report is a lossy, subjective, and sometimes absent channel; a catalogue is a historically contingent partition of a continuous pathophysiology, with boundaries that shift between editions and that fit poorly the many conditions which are combinations, prodromes, or genuinely novel (Feinstein, 1967; Loscalzo et al., 2007; Scheuermann et al., 2009). When a clinician reasons “the report resembles catalogue entry k , whose standard therapy is T_k ,” the inference is retrieval of precedent, and its validity is only as good as the catalogue’s fit to the individual in front of them.

This paper specifies a language whose execution never uses either primitive. It replaces both with a single computational act—the administration of a *characterised perturbation* to the patient and the reading of its fate—and shows that diagnosis, therapy, and the decision of when to stop all reduce to operations on that one act. The language is called *Cause-for-Concern*, written `cfc`, with source files carrying the extension `.cfc`. The name is literal: a `cfc` program does not ask “which disease?” but “is there a cause for concern, where, and by how much?”—and answers with a region of state space and a measured displacement, not a name.

1.2 The one primitive: a characterised perturbation

Let a patient’s physiology be a point x in a state space $\mathcal{M}$ (a full description of the relevant oscillator ensemble: concentrations, activities, conductances, phases). An ingested or administered compound of *known* composition is a controlled input to this system; its fate—the time-course of products and physiological responses it induces—is a function of the patient’s state. If the forward model predicting that fate is valid across the state space and not merely at a healthy reference, then the measured fate *locates* the patient in $\mathcal{M}$. This single object, the characterised perturbation, is read two ways: as a *measurement* (its fate carries information about the state) and as an *intervention* (its action displaces the state). `cfc` exposes exactly this object, under the keyword `probe`, and derives everything else from it.

1.3 Five facts, proved not assumed

The language is not an interface pasted over heuristics; its every guarantee is a theorem. Five facts organise the development, each established in the body from stated premises.

1. **The resolution floor** (Section 3). A finite observer of a physiological system cannot resolve its state to a point: there is a strictly positive floor $\beta > 0$ below which distinctions cannot be made. “The state” is therefore a *cell* of diameter $\geq \beta$, and the goal of measurement is not to name the state but to shrink the admissible region to a single cell.
2. **Diagnosis as localization** (Sections 4 and 5). Health is membership of a *coherent set* $\mathcal{M}^*$ —the states in which the oscillator ensemble is internally phase-consistent—quantified by a scalar coherence $\eta \in [0, 1]$ and a disease vector $\mathbf{D}$. A probe is *diagnostically sufficient* for

a region iff its state-to-reading map is injective down to the floor, in which case reading the probe pins the patient to one cell. No catalogue enters.

3. **The trajectory certificate** (Section 6). Because a physiological response is itself a substrate, a probe can *cascade*: a later product exists only if the earlier trajectory occurred as predicted, so the readout is a *certificate* of a path, not a snapshot. A false account cannot forge it. The correct stopping rule is *rigidity*: halt when every account other than the true one is refuted by some observed product it cannot explain.
4. **Therapy as gap-closure** (Section 7). The therapeutic target is the vector from the localized state to the nearest coherent state. Treatment is the sparsest combination of characterised interventions—drawn from a basis indexed by *effect*, not by indication—that drives that vector to the floor, solved as an ℓ_1 program and executed as a contraction whose fixed point is coherence. The disease is never named.
5. **Diagnosis and therapy are one** (Section 8). Every characterised perturbation both measures and moves; the measure–act–remeasure loop needs no external monitor and halts precisely when the state is both coherent (gap below floor) and rigidly known (no rival account survives). This double condition is the operative definition of *wellness*.

1.4 What the language contributes

On this foundation we build a full language, not a sketch. Section 2 fixes the semantic model. Sections 3 to 8 prove the five facts and their supporting lemmas. Section 9 states the one substantive demand on the runtime (that the forward model be *derived*, not fitted) as a theorem about extrapolation, because the safety of *in silico* experimentation reduces to it. Sections 10 and 11 give the lexical and context-free grammar; Section 12 the accountability type system with proofs; Section 13 the small-step operational semantics with proofs of monotonicity, determinism, and scheduler soundness; Section 14 the standard library binding each verb to a semantic operation. Section 15 gives worked programs, Section 16 the complexity of each judgment, Section 17 a constructive validation protocol requiring no clinical data, and Section 18 the reference execution model. Section 19 situates the design; Section 20 concludes.

1.5 Method and standard of rigour

Every object below is a finite-dimensional dynamical system, a finite weighted structure, or a construction on one, and every claim is stated with its hypotheses and proved from them. Where a clinical reading is offered—“cell” as differential diagnosis, “gap” as therapeutic deficit, “decline” as an honest *non liquet*—it is confined to a remark on which no theorem depends. The grammar, type system, and semantics are the language as a mathematical object; the present text, not any implementation, is normative.

2 The Semantic Model

cfc is defined over a single mathematical object—the physiological state system—and three derived constructions: the coherence functional, the cell, and the gap. We fix these before any syntax.

2.1 The state system

Definition 2.1 (Physiological state system). A *physiological state system* is a tuple $\Sigma = (\mathcal{M}, F, \Pi, \rho, \Omega)$ where:

- $\mathcal{M} \subseteq \mathbb{R}_{\geq 0}^n$ is the *state space*: a point $x \in \mathcal{M}$ is a full description of the relevant physiology (metabolite concentrations, enzyme activities, channel conductances, oscillator phases) at an instant. The dimension n is finite but may be large.
- $F: \mathcal{M} \times \Pi \rightarrow \mathbb{R}^n$ is the *dynamics*: $\dot{x} = F(x, u)$, with u an input. F is a kinetic law (mass action, Michaelis–Menten (Michaelis and Menten, 1913), or any rate law fixed by chemistry) whose parameters are, by hypothesis (Section 9), fixed by physical constants rather than fitted.
- Π is the *perturbation space*: admissible inputs (compounds of known composition and their dosing profiles).
- $\rho: \mathcal{M} \rightarrow \mathcal{Y}$ is the *readout map* into an observation space $\mathcal{Y}$: what an instrument reports of a state. The canonical readout is an abundance distribution over measurable species, $\rho(x) = (\text{abundance of species } s \text{ in state } x)_s$.
- Ω is a finite *oscillator index set*: the physiological subsystems (Section 4) whose phase-consistency defines health.

Definition 2.2 (Probe; fate; trace). A *probe* $\pi \in \Pi$ is a perturbation of *known composition*: a compound (or timed sequence) whose molecular identity and quantity are exactly specified. Given a state x_0 , the *fate* of π is the trajectory $x(t; x_0, \pi)$ obtained by integrating $\dot{x} = F(x, \pi)$ from x_0 . The *trace* is the time-indexed readout $\tau(x_0, \pi) = (\rho(x(t; x_0, \pi)))_{t \geq 0}$; a single measurement returns the reading $\rho(x(t^*))$ at a read time t^* .

Remark 2.3 (A probe is a meal of known composition). Nothing in Theorem 2.2 requires a probe to be exotic or inert. Any administered substance perturbs Σ ; a meal is such a perturbation of *unknown* composition, hence of unpredictable fate. A probe is a perturbation of the same physical class whose composition is known, and it is that knowledge—not inertness—that makes its fate a usable measurement. The perturbation is not a hazard to be avoided but the instrument itself. No theorem depends on this reading.

2.2 Two structural assumptions

We adopt two premises and carry them throughout. They are the only assumptions of the paper about the observer and the state; everything else is derived.

Assumption 1 (Finite resolution). Any physical observer of Σ has finite resolving power: there is a $\delta > 0$ such that states within distance δ produce observationally indistinguishable readouts, $\text{dist}(x, x') < \delta \Rightarrow \|\rho(x) - \rho(x')\|$ below the instrument’s discrimination threshold.

Assumption 2 (Non-exhaustibility). No finite battery of measurements exhausts the state: for every finite set of readouts there remains a distinction the system could exhibit that the battery does not resolve. Equivalently, at every scale there is a finer sub-state the current resolution collapses. This is an order condition on refinement (no terminal stage), not a claim of infinite dimension.

Remark 2.4 (Why non-exhaustibility, not infinitude). Assumption 2 is deliberately an order condition and not a cardinality claim. A living system is coupled to an environment, a history, and a regulatory context that no finite instrument battery closes off; there is always a further assay, a further time point, a further covariate. It is exactly the impossibility of *finishing* the characterisation of a state against everything it is not that forces the positive floor of Section 3, and a completed totality is neither needed nor the right object.

3 The Resolution Floor

The first quantitative fact is that resolution has a strictly positive floor, and that this floor is not a nuisance to be engineered away but the width of the region within which a state counts as

determined. It is the reason a diagnosis is a region and not a point, and the reason `cfc` returns a `Cell` rather than a label.

Definition 3.1 (Separation and residue). For a state x , its *separation* $\sigma(x)$ is the least observational effort—the discrimination distance in $\mathcal{Y}$ realised over admissible probes—required to distinguish x from every other state of $\mathcal{M}$. The *residue* $\rho_*(x) \geq 0$ is the irreducible part of this separation that no admissible measurement removes; it is 0 iff the distinction of x from the whole of $\mathcal{M}$ is completed.

Theorem 3.2 (Floor Theorem). *Under Assumptions 1 and 2 there is a constant $\beta > 0$ such that $\rho_*(x) \geq \beta$ for every state $x \in \mathcal{M}$: no state is resolved to a point.*

Proof. Suppose some state x had zero residue, $\rho_*(x) = 0$. By Theorem 3.1 this means the distinction of x from every other state of $\mathcal{M}$ is completed by a finite measurement—every state the system could exhibit has been brought to bear in separating x from it. But completing the comparison of x against all of $\mathcal{M} \setminus \{x\}$ exhausts the system’s distinctions at a finite stage, contradicting non-exhaustibility (Assumption 2). Hence $\rho_*(x) > 0$ for every x . Passing to the quotient of $\mathcal{M}$ under indistinguishability (Assumption 1, which identifies states closer than δ), the strictly positive residues are bounded below by their infimum $\beta := \inf_x \rho_*(x) > 0$: were the infimum 0 there would be states of arbitrarily small residue, i.e. nearly completed comparisons, and each in the limit would finish the comparison and exhaust the system—the same contradiction. Thus $\beta > 0$. $\square$

Corollary 3.3 (Solved is a cell). *The finest determination of a state is a cell of diameter β : the set of states an ideal measurement cannot separate from the true one. “The state” is this cell, not its centre.*

Definition 3.4 (Cell). For $x \in \mathcal{M}$, the *cell* of x is $\mathcal{C}(x) := \{x' \in \mathcal{M} : \text{dist}(x, x') < \beta\}$, the ball of radius β under the state metric. A subset $A \subseteq \mathcal{M}$ is *cell-tight* if $\text{diam}(A) < \beta$, i.e. A lies within a single cell.

Principle 3.5 (The target is the cell, not the point, and not a label). Two consequences fix the whole language. **(a)** Because the target is a cell, the diagnostic task is *localization*—shrinking the admissible region to one cell-tight set—not *recognition against a catalogue*. **(b)** Because a cell is defined intrinsically (by indistinguishability), it needs no external library of named states to be identified: a diagnosis is complete without naming a disease. `cfc`’s diagnostic output type is therefore `Cell`, never a nosological label.

Remark 3.6 (What the Floor Theorem does and does not say). Theorem 3.2 does not assert that instruments are poor; it asserts that a genuinely non-exhaustible state cannot, even with an ideal instrument, terminate at a residue-free point. A “perfect” diagnosis—one returning a point—would require completing a comparison against a state that, by hypothesis, is never completed. Sections 5 and 6 turn this into an honest stopping rule rather than a caveat.

4 Coherence: The Geometry of Health

Diagnosis needs a notion of *where health is* in $\mathcal{M}$ that does not appeal to a catalogue. We supply it through coherence: health is phase-consistency of the oscillator ensemble, a geometric property of the state, measured by a scalar in $[0, 1]$ and refined into a vector over oscillator classes.

4.1 Oscillators and the coherence index

Definition 4.1 (Oscillator; performance metric). An *oscillator* $i \in \Omega$ is a physiological subsystem with a characteristic frequency and a scalar *performance metric* $\Pi_i: \mathcal{M} \rightarrow \mathbb{R}$ (a rate, amplitude, open probability, period stability, or the like) read from the state. Each oscillator carries an

optimal value Π_i^{opt} (performance at full phase-lock) and a *degraded* value Π_i^{deg} (performance at zero phase-lock), with $\Pi_i^{\text{opt}} \neq \Pi_i^{\text{deg}}$.

Definition 4.2 (Coherence index). The *coherence index* of oscillator i at state x is the clamped affine normalisation

$$\eta_i(x) = \text{clip}_{[0,1]} \left(\frac{\Pi_i(x) - \Pi_i^{\text{deg}}}{\Pi_i^{\text{opt}} - \Pi_i^{\text{deg}}} \right), \quad \text{clip}_{[0,1]}(z) = \max(0, \min(1, z)), \quad (1)$$

so $\eta_i(x) = 1$ at optimal performance and $\eta_i(x) = 0$ at degraded performance, monotone in Π_i between them.

Proposition 4.3 (Uniqueness of the coherence normalisation). *Among affine maps $\mathbb{R} \rightarrow [0, 1]$ carrying $\Pi_i^{\text{deg}} \mapsto 0$ and $\Pi_i^{\text{opt}} \mapsto 1$, (1) is the unique one, and its clamp is the unique extension to all of $\mathbb{R}$ that is monotone, idempotent on $[0, 1]$, and constant outside.*

Proof. An affine map $z \mapsto az + b$ with two prescribed value pairs has a, b determined uniquely by solving the two linear equations, giving the ratio in (1). Extending to $\mathbb{R}$ while keeping the image in $[0, 1]$, monotone, and equal to the affine map on the pre-image of $[0, 1]$ forces the value 0 below and 1 above, which is exactly $\text{clip}_{[0,1]}$. $\square$

Definition 4.4 (Weighted cellular coherence). Given nonnegative weights $\{w_i\}_{i \in \Omega}$ with $W = \sum_i w_i > 0$, the *cellular coherence* at state x is

$$\eta(x) = \frac{1}{W} \sum_{i \in \Omega} w_i \eta_i(x) \in [0, 1]. \quad (2)$$

The *disease index* of oscillator i is $D_i(x) = 1 - \eta_i(x)$, and the *disease vector* is $\mathbf{D}(x) = (D_i(x))_{i \in \Omega} \in [0, 1]^{|\Omega|}$.

Proposition 4.5 (Coherence bounds and severity). $\eta(x) \in [0, 1]$ for every x , and the total severity $1 - \eta(x) = \frac{1}{W} \sum_i w_i D_i(x)$ is the weighted mean disease index. Both are computable in $\mathcal{O}(|\Omega|)$ from the metrics $\{\Pi_i(x)\}$.

Proof. Each $\eta_i(x) \in [0, 1]$ by the clamp; a weighted average of numbers in $[0, 1]$ with positive weights lies in $[0, 1]$. The severity identity is immediate from $D_i = 1 - \eta_i$ and linearity. Each η_i is $\mathcal{O}(1)$; summing over Ω is $\mathcal{O}(|\Omega|)$. $\square$

4.2 The coherent set

Definition 4.6 (Coherent set; health). Fix a coherence threshold $\theta \in (0, 1]$. The *coherent set* is

$$\mathcal{M}_\theta^* = \{x \in \mathcal{M} : \eta_i(x) \geq \theta \text{ for all } i \in \Omega\}, \quad (3)$$

the states in which every oscillator is within tolerance of phase-lock. A state is *healthy* iff it lies in $\mathcal{M}_\theta^*$. Where the threshold is fixed we write $\mathcal{M}^*$.

Remark 4.7 (Health is a region, not a catalogue complement). $\mathcal{M}^*$ is defined *intrinsically*, by an inequality on the state's own oscillator metrics, with no reference to any list of diseases. Disease is then simply $\mathcal{M} \setminus \mathcal{M}^*$: the states failing coherence. This is the geometric content of “disease is loss of phase-consistency,” and it is what lets diagnosis proceed by localization against $\mathcal{M}^*$ rather than by matching against named entries. No theorem depends on any clinical reading of the word “healthy.”

Lemma 4.8 (Coherence is Lipschitz where the metrics are). *If each Π_i is L_i -Lipschitz on $\mathcal{M}$, then η is L -Lipschitz with $L = \frac{1}{W} \sum_i w_i L_i / |\Pi_i^{\text{opt}} - \Pi_i^{\text{deg}}|$, and each cell $\mathcal{C}(x)$ has coherence variation at most $L\beta$.*

Proof. The clamp is 1-Lipschitz and the affine map has slope $1/|\Pi_i^{\text{opt}} - \Pi_i^{\text{deg}}|$, so η_i is $(L_i/|\Pi_i^{\text{opt}} - \Pi_i^{\text{deg}}|)$ -Lipschitz; the weighted average is L -Lipschitz by the triangle inequality. Over a cell of diameter β the coherence changes by at most $L\beta$. $\square$

5 Diagnosis as Localization

We now make precise the sense in which a probe’s reading *locates* the patient, and the exact condition under which a probe is diagnostically sufficient. Diagnosis is the shrinking of an admissible region to one cell; it succeeds when the map from state to reading loses no information above the floor.

Definition 5.1 (State-resolved forward model). A forward model of Σ is *state-resolved* if it predicts the trace $\tau(x_0, \pi)$ as a function of the initial state x_0 over all of $\mathcal{M}$, not merely at a reference x^{ref} . Write $\Phi_\pi: \mathcal{M} \rightarrow \mathcal{Y}$, $\Phi_\pi(x_0) = \rho(x(t^*; x_0, \pi))$, for the predicted reading map of probe π at read time t^* .

Remark 5.2 (Why a reference-only model is insufficient). A model valid only at x^{ref} predicts one reading; a measured deviation certifies only *that* the patient differs, not *how*. Inversion—reading the state off the reading—requires the reading to vary with the state, i.e. Φ_π as a genuine map of x_0 . This is the crux of localization and the sole substantive demand the method places on the model (discharged in Section 9).

Definition 5.3 (Localization map). Given probe π and a measured reading $y \in \mathcal{Y}$ at time t^* , the *localization* induced by π is the preimage

$$\Lambda_\pi(y) = \{x_0 \in \mathcal{M} : \|\Phi_\pi(x_0) - y\| < \delta\}, \quad (4)$$

the set of states whose predicted reading matches the measurement to within the resolution δ (Assumption 1). Reading the probe replaces the prior admissible region A by $A \cap \Lambda_\pi(y)$.

Theorem 5.4 (Localization Theorem / diagnostic sufficiency). *A probe π is diagnostically sufficient for a region $A \subseteq \mathcal{M}$ if and only if its reading map restricted to A is injective down to the floor: for all $x_0, x'_0 \in A$,*

$$\|\Phi_\pi(x_0) - \Phi_\pi(x'_0)\| < \delta \implies \text{dist}(x_0, x'_0) < \beta. \quad (5)$$

Under this condition, for every measurement y the localization $\Lambda_\pi(y) \cap A$ is cell-tight, so the probe pins the state to one cell. Conversely, if (5) fails, there are two states at distance $\geq \beta$ producing the same reading within δ , and no single reading of π separates their cells.

Proof. ($\Leftarrow$) Assume (5). Take $x_0, x'_0 \in \Lambda_\pi(y) \cap A$. Then $\|\Phi_\pi(x_0) - y\| < \delta$ and $\|\Phi_\pi(x'_0) - y\| < \delta$, so $\|\Phi_\pi(x_0) - \Phi_\pi(x'_0)\| < 2\delta$; applying (5) with the instrument threshold set at 2δ (or, equivalently, absorbing the factor into δ , as Assumption 1 tolerates a single fixed discrimination scale) yields $\text{dist}(x_0, x'_0) < \beta$. Hence any two states in the localization are within β , so $\text{diam}(\Lambda_\pi(y) \cap A) < \beta$: the set is cell-tight. ($\Rightarrow$) Suppose (5) fails: there exist $x_0, x'_0 \in A$ with $\|\Phi_\pi(x_0) - \Phi_\pi(x'_0)\| < \delta$ yet $\text{dist}(x_0, x'_0) \geq \beta$. Set $y = \Phi_\pi(x_0)$. Then both states lie in $\Lambda_\pi(y)$ (the second because its reading is within δ of y), so the localization contains two points at distance $\geq \beta$, meeting two distinct cells; no single reading distinguishes them, and π is insufficient for A . $\square$

Corollary 5.5 (Probe design is an injectivity problem). *Designing a diagnostic probe (or panel, Section 6) for a region A is choosing a probe whose reading map is floor-injective on A . Diagnosis is then evaluation of Λ_π at the measured reading. No disease library enters; the output is the cell (Theorem 3.5).*

Corollary 5.6 (Coherence is read from the cell). *If π is diagnostically sufficient for A and each metric Π_i is L_i -Lipschitz, then reading π determines every coherence index η_i and hence the disease vector $\mathbf{D}$ to within $L_i\beta$ (Theorem 4.8): the localized cell fixes the health geometry up to the floor.*

Proof. By Theorem 5.4 the localization is cell-tight, of diameter $< \beta$; by Theorem 4.8 each η_i varies by at most $L_i\beta$ over a cell, so its value is fixed to that tolerance, and $\mathbf{D}$ with it. $\square$

6 Cascading Probes and the Trajectory Certificate

A single reading of a single probe may fail to be floor-injective; more information is needed. Rather than administer many independent probes, we exploit that each physiological product is itself a reactive species, so a probe can generate its own successors—a *cascade*—whose structure yields both richer localization and a strong non-forgability property that a mere agreement of magnitudes cannot provide.

Definition 6.1 (Cascade probe; panel). A *cascade probe* is a probe π whose fate passes through a designed sequence of species $\pi = P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow \dots$, where each P_{k+1} is a product of a reaction consuming P_k , and each transition $P_k \rightarrow P_{k+1}$ is *state-gated*: whether P_{k+1} forms in observable quantity within the read window depends on the state x . A *panel* is a finite family of probes (cascade or single) read together.

Definition 6.2 (Traversed path; committed record). The *traversed path* of a cascade in state x_0 is the maximal prefix $P_0 \rightarrow \dots \rightarrow P_{d(x_0)}$ of state-gated transitions proceeding to observable extent; $d(x_0)$ is the *depth*. Because each transition consumes its predecessor and produces its successor, the species population is a *monotone committed record*: a later species' presence is not undone by subsequent dynamics on the read timescale, so the reading reports the path, not only the instantaneous state.

Theorem 6.3 (Certificate Theorem). *Let P_k be a species that forms in observable quantity only if every transition $P_0 \rightarrow \dots \rightarrow P_k$ has occurred. Then observation of P_k entails that the state x_0 enables the whole path: the set of states consistent with the reading is contained in*

$$\bigcap_{j \leq k} \{x : \text{transition } j \text{ is state-enabled at } x\}. \quad (6)$$

Consequently, a hypothesised state $\hat{x}$ that fails to enable some transition $j \leq k$ is refuted by the presence of P_k : no such $\hat{x}$ can reproduce the reading.

Proof. The forward implication is the existence condition of Theorem 6.1: P_k observable $\Rightarrow$ each prior transition proceeded $\Rightarrow$ each prior transition was state-enabled at x_0 . Hence the consistent-state set is the intersection of the enabling sets of transitions $1, \dots, k$, which is (6). If $\hat{x}$ lies outside the enabling set of some $j \leq k$, then at $\hat{x}$ transition j does not proceed, so P_j —and therefore P_k , which requires P_j —is absent; observing P_k contradicts $\hat{x}$. $\square$

Principle 6.4 (Existence beats consistency). Theorem 6.3 is a stronger epistemic guarantee than agreement of magnitudes. A consistency check asks whether a hypothesis *matches* observed values; it can be satisfied by a well-chosen falsehood. A cascade reading asks whether the observed species *could have formed*; a falsehood failing to enable a transition does not merely mismatch a number—it predicts a species that is *absent*, and absence cannot be argued around. The terminal products are a constructive proof of their own prerequisites.

Theorem 6.5 (Rigidity: the correct halting criterion). *Let a panel yield observed species $S = \{P_k\}$, and define the surviving set*

$$\mathcal{M}_S = \bigcap_{P_k \in S} \{x : \text{the path to } P_k \text{ is enabled at } x\}. \quad (7)$$

Then $\mathcal{M}_S$ is non-increasing as S grows, and the state is sufficiently localized exactly when $\mathcal{M}_S$ is cell-tight ($\text{diam}(\mathcal{M}_S) < \beta$): every account of the state outside a single cell is refuted by the absence it would predict of some observed species. This is a rigidity condition—no alternative account survives—not a threshold on any scalar.

Proof. Each observed species adds a constraint $\{x : \text{its path enabled}\}$; $\mathcal{M}_S$ is the intersection of these constraints, and an intersection is non-increasing under the addition of sets, so $\mathcal{M}_S$ shrinks (weakly) as S grows. When $\text{diam}(\mathcal{M}_S) < \beta$, $\mathcal{M}_S$ lies within one cell (Theorem 3.3); any x outside that cell violates at least one constraint, i.e. predicts some observed P_k absent, and is refuted by Theorem 6.3. Conversely, if $\text{diam}(\mathcal{M}_S) \geq \beta$, two states in $\mathcal{M}_S$ occupy distinct cells and both reproduce the reading, so the state is not localized. Sufficiency is therefore exactly cell-tightness. $\square$

Corollary 6.6 (Single-measurement richness). *A readout returning the full species distribution—the whole population of cascade products present, not one terminal product—delivers the entire constraint family S in one measurement, hence $\mathcal{M}_S$ in one shot. The instrument’s report is the committed record of the traversed path.*

Remark 6.7 (A continuously working probe). Because a cascade is self-propagating and its record monotone, a single administration installs a process that keeps advancing and recording after administration. The state is therefore readable at any later time by any observer with the instrument: the probe does not deliver a value once and expire; it remains a live, queryable source until the cascade completes. “Continuous” is here a property of the probe—always working—not of the measurement schedule. No theorem depends on this reading.

7 Therapy as Gap-Closure Without a Named Disease

We now define treatment purely in terms of the localized state and coherence, with no reference to a disease label. Therapy closes the measured gap to the coherent set by the sparsest characterised interventions, executed as a contraction.

Definition 7.1 (Gap). For a localized state x , the *gap* is the displacement to the nearest coherent state,

$$\mathbf{g}(x) = x^* - x, \quad x^* = \arg \min_{z \in \mathcal{M}^*} \text{dist}(x, z), \quad (8)$$

a measured vector defined without naming what caused the deviation. (If the minimiser is non-unique, fix any measurable selection; no theorem depends on the choice.)

Definition 7.2 (Intervention basis). An *intervention basis* $\mathcal{B} = \{u_1, \dots, u_m\}$ is a collection of characterised compounds, each with a known state-resolved effect: applying u_i at state x produces a local displacement $e_i(x) \in \mathbb{R}^n$, the compound’s effect at the current state computed by the forward model. Compounds are indexed by their displacement vectors (*properties*), not by indication.

Remark 7.3 (Repurposing is the default, not the exception). Because a compound enters therapy through its displacement $e_i(x)$ and not through any indication, any compound whose displacement has a component along the gap is admissible for any patient. The distinction “on-label / off-label” does not arise: there are no labels, only properties and gaps, and the pharmacopeia is addressable by its effects (Pushpakom et al., 2019). No theorem depends on this reading.

Problem 7.4 (Minimal gap-closing intervention). Given a localized state x with gap $\mathbf{g}(x)$ and basis $\mathcal{B}$, choose nonnegative doses $c \in \mathbb{R}_{\geq 0}^m$ to close the gap with the fewest compounds:

$$\min_{c \in \mathbb{R}_{\geq 0}^m} \|c\|_1 \quad \text{s.t.} \quad \left\| \mathbf{g}(x) - \sum_i c_i e_i(x) \right\| \leq \beta. \quad (9)$$

The ℓ_1 objective selects a sparse intervention set; the constraint requires the combined local displacement to close the gap to within the floor.

Proposition 7.5 (Well-posedness and sparsity). *Theorem 7.4 is a convex program; it is feasible iff $\mathbf{g}(x)$ lies within β of the cone $\text{cone}\{e_i(x)\}$, and when feasible it attains a minimiser. If the vectors $\{e_i(x)\}$ are in general position, a minimiser is supported on at most n compounds.*

Proof. The objective $\|c\|_1$ is convex and the constraint set $\{c \geq 0 : \|\mathbf{g} - \sum_i c_i e_i\| \leq \beta\}$ is the intersection of the nonnegative orthant with the preimage of a closed ball under an affine map, hence closed and convex; the problem is convex (Boyd and Vandenberghe, 2004). Feasibility is exactly the stated cone condition by definition of the constraint. On the feasible set the objective is coercive on the orthant (grows with $\|c\|_1$) and continuous, and the feasible set is closed, so a minimiser is attained. Sparsity: the feasible set is a polyhedron intersected with a ball; by the representation of basic feasible solutions of the linear relaxation, and the sparsity-inducing geometry of the ℓ_1 ball (Candès et al., 2006; Tibshirani, 1996), a minimiser lies on a face of dimension $\leq n$, so at most n doses are nonzero when the $e_i(x)$ are in general position. $\square$

The local displacements $e_i(x)$ are exact only at x ; applying the doses moves the state, after which the displacements change. Therapy is therefore iterative.

Definition 7.6 (Therapeutic map; contraction). Let $T: \mathcal{M} \rightarrow \mathcal{M}$ send a state to the state resulting from localizing it (Section 5), solving Theorem 7.4, and applying the chosen intervention. T is a *gap contraction* on a neighbourhood U of $\mathcal{M}^*$ if there is $\lambda \in [0, 1)$ with $\text{dist}(Tx, \mathcal{M}^*) \leq \lambda \text{dist}(x, \mathcal{M}^*)$ for all $x \in U$.

Theorem 7.7 (Convergence to coherence). *If T is a gap contraction on U with modulus λ , then iterating T from any $x_0 \in U$ drives the state to the coherent set: the sequence $x_{k+1} = Tx_k$ satisfies $\text{dist}(x_k, \mathcal{M}^*) \leq \lambda^k \text{dist}(x_0, \mathcal{M}^*) \rightarrow 0$, and reaches $\text{dist}(x_k, \mathcal{M}^*) \leq \beta$ after at most $k^* = \lceil \log_{1/\lambda}(\text{dist}(x_0, \mathcal{M}^*)/\beta) \rceil$ steps, at which point the floor forbids further resolvable contraction. No global linearity of drug effects is required—only that each measured, locally-composed step contracts the gap.*

Proof. By the contraction inequality and induction, $\text{dist}(x_k, \mathcal{M}^*) \leq \lambda^k \text{dist}(x_0, \mathcal{M}^*)$; since $\lambda \in [0, 1)$ this tends to 0. Solving $\lambda^k \text{dist}(x_0, \mathcal{M}^*) \leq \beta$ for k gives $k \geq \log_{1/\lambda}(\text{dist}(x_0, \mathcal{M}^*)/\beta)$, i.e. the stated k^* . Below β the states are indistinguishable (Assumption 1), so no further contraction is resolvable and the iteration halts. This is the standard fixed-point iteration of a contraction (Banach, 1922), specialised to distance-to-a-set with the floor as halting threshold. $\square$

Principle 7.8 (Cure without naming). Theorem 7.7 restores coherence—closes the gap—without ever identifying a disease. The therapeutic objective is the measured residual gap, and success is its reduction to the floor. Naming, if desired, is a downstream description of which cell the state occupied; it is never a prerequisite for treatment.

8 Diagnosis and Therapy Are One Operation

The measurement primitive and the therapeutic primitive are the same object read in two directions. This is what lets `cfc` expose a single verb for both and makes the treatment loop self-monitoring.

Theorem 8.1 (Measurement–action identity). *Every characterised perturbation $u \in \Pi$ acts simultaneously as a measurement and as an intervention:*

- (i) *as a measurement, its trace $\tau(x, u)$ localizes the state (Theorem 5.4);*
- (ii) *as an intervention, its displacement moves the state (Theorem 7.2);*

and these are not two applications but one: applying u yields a trace (information) by moving the state (action). A probe is a compound read for its fate; a drug is a compound applied for its displacement; they are the same primitive, and each does both.

Proof. A perturbation induces a single trajectory (Theorem 2.2). Its readout is the measurement (i); its endpoint is the action (ii). Both are functions of the one applied u : the map $u \mapsto (\text{trace}, \text{displacement})$ has one input and two outputs computed from the same induced trajectory. $\square$

Corollary 8.2 (Self-terminating therapeutic loop). *Because the therapeutic step is also a measurement, the contraction of Section 7 needs no external monitor: each applied intervention re-localizes the state as a side effect of acting. The loop measure–act–remeasure is a single iterated perturbation, and it halts by the rigidity criterion (Theorem 6.5)—when the accumulated record over-determines the state to within the floor—which is simultaneously the diagnostic sufficiency condition and the therapeutic success condition.*

Theorem 8.3 (Wellness Theorem). *Combining Theorems 6.5 and 7.7, the terminal condition of the cfc loop is a state that is both coherent (its gap is below the floor) and rigidly known (no alternative account survives its own probes). This double condition—and only it—is the operative definition of wellness in the calculus: a well system returns its probes at full resolution and admits no gap-closing intervention beyond the floor.*

Proof. Termination of the loop requires both sub-conditions. If the gap exceeds the floor, Theorem 7.7 supplies a further contracting step, so the loop continues; hence at termination $\text{dist}(x, \mathcal{M}^*) \leq \beta$ (coherent). If more than one cell survives, Theorem 6.5 shows a further probe can refute a rival account, so the loop continues; hence at termination the surviving set is cell-tight (rigidly known). Conversely, when both hold, no probe changes the localized cell and no intervention resolvably reduces the gap, so the loop is at a fixed point. The two conditions are thus jointly necessary and sufficient for termination. $\square$

Remark 8.4 (Concern as the negation of wellness). The language’s name is the negation of Theorem 8.3: a *cause for concern* is any surviving cell whose coherence falls below threshold, i.e. any x in the localized set with $\eta(x) < \theta$. A cfc program’s **concern** value carries exactly this—the localized cell together with its coherence deficit and disease vector—never a disease name. No theorem depends on the reading.

9 The Runtime Must Be Derived, Not Fitted

Every construction above presupposes a state-resolved forward model (Theorem 5.1). We now state the one condition this model must meet, as a theorem about extrapolation, because the safety of *in silico* experimentation—the whole point of running a therapy search against a model before touching a patient—reduces to it.

Definition 9.1 (Fitted vs. derived model). A forward model is *fitted* if its rate parameters are estimated to minimise error against observations drawn from a training region $A_{\text{tr}} \subseteq \mathcal{M}$. It is *derived* if its rate laws and their parameters are fixed by physical law (thermodynamic and kinetic constants, conservation constraints) independently of any observation region (Orth et al., 2010; Palsson, 2015).

Theorem 9.2 (Extrapolation gap of fitted models). *A fitted model carries no guarantee of accuracy outside A_{tr} : there exist state systems agreeing with the training data on A_{tr} to any tolerance yet disagreeing arbitrarily on $\mathcal{M} \setminus A_{\text{tr}}$. Consequently a fitted model’s reading map Φ_π is trustworthy only near A_{tr} , whereas a derived model, whose F is fixed pointwise on $\mathcal{M}$ by law, admits a uniform error bound over $\mathcal{M}$.*

Proof. Standard non-identifiability. The training data constrain F only on A_{tr} : any extension of F agreeing there is admissible, and two admissible extensions may be chosen to diverge by any prescribed amount off A_{tr} (add to one a smooth bump supported in $\mathcal{M} \setminus A_{\text{tr}}$ of arbitrary height).

Hence the fitted Φ_π , obtained by integrating F , is pinned only near A_{tr} . A derived F is not an extension chosen to fit but a function specified on all of $\mathcal{M}$ by its physical form, so its accuracy is governed by the fidelity of the physical law, uniformly, not by proximity to data. $\square$

Principle 9.3 (Free exploration requires a derived runtime). The value of `cfc` is that a clinician may experiment freely against the model—searching the intervention basis, testing cascades, exploring the gap—at no risk to the patient, applying to the patient only the intervention the search selects. But free exploration *is* off-training-region evaluation by definition. By Theorem 9.2 a fitted runtime becomes unreliable exactly where the search explores; only a derived runtime supports risk-free *in silico* search. The requirement is not stylistic: the safety of the method reduces to the extrapolation validity of the runtime. A `cfc` runtime is *sound* only if its forward model is derived (Theorem 14.4).

10 Lexical Structure

A `cfc` source file is UTF-8 text with extension `.cfc`. Lexing produces a token stream from the classes below; tokenisation follows maximal munch, with keywords taking priority over identifiers (Hopcroft et al., 2006).

Definition 10.1 (Token classes).

- **Comments.** `-` to end of line; discarded, lexically equivalent to whitespace.
- **Keywords.** `floor`, `patient`, `probe`, `panel`, `measure`, `localize`, `cell`, `gap`, `close`, `toward`, `coherent`, `using`, `basis`, `until`, `resolved`, `contested`, `decline`, `concern`, `when`, `emit`, `assert`, `yield`, `let`, `in`, `as`, `with`, `import`, `module`, `export`, `monitor`, `every`, `observe`, `against`, `reference`.
- **Identifiers.** `[A-Za-z_][A-Za-z0-9_]*`, excluding keywords.
- **Numbers.** decimal integers and floats with optional exponent and optional floor annotation (Theorem 10.2).
- **Strings.** double-quoted, with the usual escapes.
- **Operators and punctuation.** `:=` (bind), `»` (cascade, sequential composition of probe stages), `||` (parallel panel dispatch), `( ) { } [ ] , : . > < >= <= ==` and the field-access operator `.`.

Whitespace is insignificant except as a token separator; `cfc` is not layout-sensitive.

Notation 10.2 (Floor-annotated literals). A numeric literal may carry an explicit floor annotation with the suffix form `value#floor` (read “value at floor”); e.g. `0.82#1e-3`. A literal without an explicit floor inherits the ambient program floor set by the most recent `floor` declaration in scope. There is no way to write a literal of zero floor (Theorem 12.4).

11 Context-Free Grammar

We give the grammar in extended Backus–Naur form (Aho et al., 2006; Backus, 1959; Naur, 1963). Nonterminals are *italicised*; terminals are **typewriter**; `{x}` is zero-or-more repetition, `[x]` optional, `|` alternation. The start symbol is *program*.

Definition 11.1 (cfc grammar).

```

program ::= { decl }
decl ::= floorDecl | import | moduleDecl | basisDecl | patientDecl
floorDecl ::= floor number
import ::= import qname
moduleDecl ::= module ident { { decl } }
basisDecl ::= basis ident { { probeDef } }
probeDef ::= probe ident { spec }
spec ::= reference: expr observe: stageChain
patientDecl ::= patient ident { { stmt } }
stmt ::= bind | measureStmt | diagnoseStmt | treatStmt
       | monitorStmt | assertStmt | expr
bind ::= [let] ident := expr
measureStmt ::= measure panelRef yield ident
diagnoseStmt ::= localize ident
               using panelRef
               until admit
               yield ident
treatStmt ::= close gap of ident
            toward coherent
            using basis ident
            until admit
            yield ident
monitorStmt ::= monitor ident every number { { stmt } }
admit ::= resolved [otherwise decline]
       | coherent [otherwise decline]
       | cond
panelRef ::= panel { stageChain } | qname
stageChain ::= probeRef { » probeRef }
            | stageChain || stageChain
probeRef ::= qname ( [argList] )
assertStmt ::= assert cond [emit string]
cond ::= expr relop expr
relop ::= > | < | >= | <= | ==
argList ::= arg { , arg }
arg ::= [ident :] expr
qname ::= ident { . ident }
expr ::= ident | number | string | probeRef | expr.ident | ( expr )
type ::= Cell | Coherence | DiseaseVector | Gap | Probe | Panel
       | Intervention | Reading | Verdict | Concern

```


Remark 11.2 (Reading the core forms). A `cfc` program has three verbs, one per operational theorem. `measure` administers a panel and binds its `Reading` (Theorem 2.2). `localize ...until resolved` runs the rigidity loop of Theorem 6.5, binding a `Cell`; its optional `otherwise decline` names the contested-closure outcome (Theorem 13.8). `close gap ...toward coherent until coherent` runs the gap-closing contraction of Theorem 7.7, binding an `Intervention`; `coherent` means “gap below the floor,” not a numeric threshold. `monitor ...every  $\Delta t$` re-runs its body on a schedule, realising the continuously working probe of Theorem 6.7. `assert` is a pre-flight check that may halt with a message. A `patient` block is the unit of orchestration: a sequence of these statements whose yields may feed later ones.

Theorem 11.3 (Unambiguity of the core grammar). *The grammar of Theorem 11.1 is unambiguous, and is $LL(1)$ after left-factoring the `stmt` and `admit` alternatives, each of which is selected by a distinct leading keyword.*

Proof. Every statement form begins with a distinct keyword (`measure`, `localize`, `close`, `monitor`, `assert`, `let`) or is an *expr*; the leading token thus selects the production, giving $LL(1)$ selection with one token of lookahead. The *admit* alternatives are distinguished by their leading keyword (`resolved`, `coherent`, or an *expr* beginning a *cond*). The optional `otherwise decline` is bound to its enclosing *admit* by the nearest enclosing rule, with no dangling ambiguity because each *admit* occurs in exactly one statement. The expression grammar is a standard member/call-suffix cascade, unambiguous by the usual argument (Aho et al., 2006). Hence every sentential form has a unique derivation. $\square$

12 The Accountability Type System

`cfc`’s types record *what kind of clinical object* a value is and *at what floor* it was determined. The system’s purpose is to make two properties statically checkable: that no value claims a resolution finer than the floor (an accountability property), and that no therapy is declared *cured* without a localized cell to have closed the gap of.

12.1 Types and judgements

Definition 12.1 (Types). The `cfc` types are

$\tau ::= \text{Reading} \mid \text{Cell} \mid \text{Coherence} \mid \text{DiseaseVector} \mid \text{Gap} \mid \text{Probe} \mid \text{Panel} \mid \text{Intervention} \mid \text{Verdict}$

`Reading` is a raw measured readout (Theorem 2.2); `Cell` is a localized cell-tight region (Theorem 3.4); `Coherence` $\in [0, 1]$ and `DiseaseVector` $\in [0, 1]^{|\Omega|}$ are the health scalars/vectors (Theorem 4.4); `Gap` is a measured displacement (Theorem 7.1); `Intervention` is a chosen dose vector with its predicted displacement; `Verdict` is the outcome of a rigidity or coherence admission (`resolved/contested/coherent`); `Concern` bundles a cell, its coherence, and its disease vector (Theorem 8.4). Every value carries a *floor annotation* $\beta(v) > 0$ and, where it denotes a determined region, a *residue* $\rho_*(v) \geq \beta(v)$.

Definition 12.2 (Typing judgement). A judgement $\Gamma \vdash e : \tau @ \beta$ reads: “in floor context Γ (fixing the ambient floor and the types of bound identifiers), expression e has type τ and, if it denotes a determined region, does so at floor $\beta > 0$.” The context carries the ambient floor set by the most recent `floor` declaration and the types of bound identifiers.

12.2 The floor-positivity rule

Rule 12.3 (Floor positivity). Every value-introducing rule carries the side condition $\beta > 0$ and, for a region-denoting value, $\rho_* \geq \beta$. A literal or expression whose annotation forces $\beta \leq 0$ or

$\rho_* < \beta$ is ill-typed. In particular,

$$\frac{\Gamma \vdash e : \text{Panel} \quad \Gamma \vdash r : \text{Reading} \quad \beta_\Gamma > 0}{\Gamma \vdash \text{localize } \dots \text{ using } e \text{ until resolved } \dots : \text{Cell} @ \beta_\Gamma}$$

where β_Γ is the ambient floor set by the most recent `floor` declaration.

Theorem 12.4 (No zero-residue value). *There is no well-typed cfc expression of residue 0: every typeable region-denoting value (a `Cell`, `Gap`, or `Concern`) has $\rho_* \geq \beta > 0$. Equivalently, a point-resolved diagnosis is not expressible.*

Proof. By induction on typing derivations. The only region-introducing rules are the `localize` rule (Theorem 12.3, side condition $\beta_\Gamma > 0$, whose result carries residue = $\text{diam}(\mathcal{M}_S)$ which is $\geq \beta$ by Theorem 6.5 at admission), the `close/gap` rules (whose gap residue is the post-contraction distance, $\geq \beta$ by Theorem 7.7), and `Concern` construction (which packages a `Cell` and inherits its residue). Floor-annotated literals carry residue equal to their positive floor by Theorem 10.2. No rule introduces a value of residue $< \beta$, and $\beta > 0$ is a side condition on every floor introduction; no rule lowers the residue of an already-typed value. Hence no derivation concludes $\rho_* = 0$. $\square$

12.3 The cell-before-cure rule

Rule 12.5 (Cell before cure). A `close gap` statement type-checks only if its subject is a `Cell` (a localized state), never a raw `Reading` or unlocalized region:

$$\frac{\Gamma \vdash x : \text{Cell} @ \beta \quad \Gamma \vdash b : \text{Panel} \text{ (basis)}}{\Gamma \vdash \text{close gap of } x \text{ toward coherent using basis } b \dots : \text{Intervention} @ \beta}$$

A `close` whose subject is not of type `Cell` is rejected at compile time.

Theorem 12.6 (No therapy without localization). *No well-typed cfc program applies a gap-closing intervention to a state that has not first been localized to a cell. Equivalently, the gap $g(x)$ of Theorem 7.1 is well-defined at every typed `close`, because x is a `Cell`.*

Proof. By Theorem 12.5 the only rule producing an `Intervention` from a `close` requires its subject to have type `Cell`. A `Cell` is introduced only by the `localize` rule (Theorem 12.3), which runs the rigidity loop to cell-tightness. Hence any typed `close` acts on a localized cell, and g (defined via the nearest coherent state to a state, Theorem 7.1) is well-defined on it. A `close` on a `Reading` or other non-`Cell` value has no typing derivation. $\square$

Theorem 12.7 (Type soundness). *cfc typing is sound for the operational semantics of Section 13: if $\Gamma \vdash e : \tau @ \beta$ and e evaluates to a value v , then v has type τ , floor $\beta > 0$, and (if τ is region-denoting) residue $\rho_*(v) \geq \beta$.*

Proof sketch. Standard syntactic soundness by progress and preservation (Pierce, 2002; Wright and Felleisen, 1994). *Progress:* a well-typed closed expression is a value or takes a step, by inspection of the reduction rules of Section 13, each of which has a syntactically disjoint premise from every other rule of the same constructor. *Preservation:* each reduction preserves the static type and floor annotation of Theorem 12.3, and, by Theorem 12.4 applied at every step, preserves $\rho_* \geq \beta > 0$ for every region value. The only non-standard element is that reduction is state-mutating (Section 13: evaluating `measure/localize/close` commits measurements and advances the committed-measurement count); preservation still holds because the count is monotone (Theorem 13.2), so no reduction retroactively lowers a previously-established residue below its floor. Soundness of the runtime additionally requires a derived forward model (Theorem 14.4); with a fitted model the type system remains sound as a syntactic discipline but the values' correspondence to the patient is only local (Theorem 9.2). $\square$

13 Operational Semantics

We give a small-step semantics (Plotkin, 2004) over configurations $\langle \Sigma, M, e \rangle$, where Σ is the current state system with its accumulated committed measurements, $M \in \mathbb{N}$ is the *committed-measurement count* (the program clock), and e is the expression under evaluation. We write $\langle \Sigma, M, e \rangle \longrightarrow \langle \Sigma', M', e' \rangle$ for one step. Evaluation *is* measurement: a `measure/localize/close` step commits measurements into Σ and advances M .

13.1 Committed measurements and the intrinsic clock

Definition 13.1 (Committed measurement; committed record). A *committed measurement* is a reduction that administers one probe and records its reading against Σ (refining the surviving set $\mathcal{M}_S$ or applying one contraction step). The *committed record* $M \in \mathbb{N}$ counts committed measurements taken so far in the current evaluation.

Theorem 13.2 (Monotonicity of the committed record). *Along any evaluation, M is non-decreasing and strictly increases at every committed measurement. No reduction decrements M ; re-evaluating a probe already administered is a new committed measurement at a strictly higher M , never a cached recomputation.*

Proof. Every rule administering a probe (Theorems 13.4 to 13.6 below) increments M by the number of probes it commits, which is ≥ 1 . No rule of Section 13 removes a committed measurement or decrements M : an “undo” (e.g. a compensating counter-probe) is itself a further committed measurement, incrementing M . Hence M is monotone and strictly increases per committed measurement, and two configurations differing only in M are distinct states. $\square$

Corollary 13.3 (Measurement, not simulation). *Because evaluation mutates Σ and advances M , a `cfc` probe cannot be a pure function memoised across calls: each occurrence performs the measurement afresh and is recorded in the clock. Running the program is performing the measurements it names.*

13.2 Reduction rules

Rule 13.4 (Measure commits and counts).

$$\frac{y = \text{ADMINISTER}(\Sigma, \Pi_{\text{panel}})}{\langle \Sigma, M, \text{measure } p \text{ yield } r \rangle \longrightarrow \langle \Sigma \oplus y, M + |p|, v_{\text{Reading}}(y) \rangle}$$

where `ADMINISTER` integrates the panel’s probes against the model and returns the reading y , $\Sigma \oplus y$ records it, and $|p|$ is the number of probes in the panel.

Rule 13.5 (Localize converges).

$$\frac{\mathcal{M}_S = \text{SURVIVE}(\Sigma, p) \quad \text{diam}(\mathcal{M}_S) < \beta \quad |\text{cells of } \mathcal{M}_S| = 1}{\langle \Sigma, M, \text{localize } x \text{ using } p \text{ until resolved yield } c \rangle \longrightarrow \langle \Sigma', M + k, v_{\text{Cell}}(\mathcal{M}_S) \rangle}$$

where `SURVIVE` accumulates k probe readings into the surviving set of Theorem 6.5 and v_{Cell} is the resulting accountable cell. If instead $\mathcal{M}_S$ stabilises with more than one cell, the `otherwise decline` branch fires:

$$\frac{\text{SURVIVE}(\Sigma, p) = \mathcal{M}_S \text{ stable, } |\text{cells}| > 1}{\langle \Sigma, M, \text{localize } x \text{ using } p \text{ until resolved otherwise decline yield } c \rangle \longrightarrow \langle \Sigma', M + k, v_{\text{decline}}(\mathcal{M}_S) \rangle}$$

Rule 13.6 (Close contracts to coherence).

$$c^* = \text{SOLVE}(g(x), \mathcal{B}_b) \quad x' = x + \sum_i c_i^* e_i(x) \\ \text{dist}(x', \mathcal{M}^*) \leq \beta$$

$\langle \Sigma, M, \text{close gap of } x \text{ toward coherent using basis } b \text{ until coherent yield } t \rangle \longrightarrow \langle \Sigma', M + 1, v_{\text{Intervention}} \rangle$

where SOLVE solves Theorem 7.4. If $\text{dist}(x', \mathcal{M}^*) > \beta$ the state is re-localized and the rule re-fires on x' (one contraction step per firing), realising the iteration of Theorem 7.7; the **otherwise decline** branch fires if SOLVE is infeasible (Theorem 7.5).

Theorem 13.7 (Determinism). *For a fixed initial configuration and a fixed forward model and basis, cfc evaluation has at most one normal form: if $\langle \Sigma, M, e \rangle \longrightarrow^* \langle \cdot, \cdot, v_1 \rangle$ and $\langle \Sigma, M, e \rangle \longrightarrow^* \langle \cdot, \cdot, v_2 \rangle$ then $v_1 = v_2$.*

Proof. ADMINISTER , SURVIVE , and SOLVE are total, deterministic functions of Σ and the syntactic clauses (integration of a fixed model, intersection of enabling sets, and a convex program with a fixed tie-break, respectively). A **localize** terminates in exactly one of the two mutually exclusive states of Theorem 13.5, determined by whether $\mathcal{M}_S$ stabilises at one cell or more—a fact about the fixed model, not a runtime choice. Two cell-tight surviving sets containing the state are observationally identical (same cell), so the representative cell is unique. A **close** likewise deterministically produces the unique minimal- ℓ_1 displacement under the fixed tie-break, or declines on infeasibility. Hence the normal form is unique. $\square$

Theorem 13.8 (Convergent closure or honest decline). *Every localize over a panel with finitely many registered probes terminates in exactly one of two states: (i) resolved — $\mathcal{M}_S$ stabilises to a single cell, reported as a **Cell**; or (ii) contested — $\mathcal{M}_S$ stabilises but contains more than one cell, reported as a **decline** carrying the distinct surviving cells. No localize over a finite panel runs forever without reaching one of these.*

Proof. The registered panel is finite, so the reachable set of readings is finite and $\mathcal{M}_S$ (an intersection of finitely many enabling constraints) can shrink by only finitely many steps before every remaining probe has been tried; at that point $\mathcal{M}_S$ is stable (rigidity by exhaustion, Theorem 6.5). If the stable $\mathcal{M}_S$ is cell-tight, case (i) holds; else it meets > 1 cell and, by Theorem 3.3, no single cell is licensed, so case (ii) holds and the program reports the contested cells. The cases are exhaustive and mutually exclusive on a non-empty surviving set. $\square$

Remark 13.9 (Decline is not failure). Theorem 13.8(ii) formalises the requirement that a clinical runtime be able to report “the evidence does not single out one state” as a first-class, correctly-terminating outcome—an honest *non liquet*—rather than looping or silently emitting one contested cell as if unique. A contested closure is itself useful output: it identifies precisely which cells remain and what further probe would separate them.

Theorem 13.10 (Scheduler soundness). *The cfc scheduler never re-probes a cell already resolved to cell-tightness, and never starves a convergent close: every gap-contracting step that reduces $\text{dist}(\cdot, \mathcal{M}^*)$ by a fixed factor is eventually taken, so a contraction (Theorem 7.6) reaches $\text{dist} \leq \beta$ in finitely many steps.*

Proof. By Theorem 13.2 the committed record is monotone, so a resolved cell (recorded in Σ) is not revisited: re-probing it would strictly increase M without shrinking the already-cell-tight $\mathcal{M}_S$, and the scheduler, which advances only on strict progress in $\mathcal{M}_S$ or $\text{dist}(\cdot, \mathcal{M}^*)$, does not schedule such a step. For non-starvation: each **close** firing applies one contraction step reducing $\text{dist}(\cdot, \mathcal{M}^*)$ by a factor $\leq \lambda < 1$ (Theorem 7.6); by Theorem 7.7 the distance reaches $\leq \beta$ after $k^* = \lceil \log_{1/\lambda}(\text{dist}_0 / \beta) \rceil$ firings, a finite bound, so the convergent therapy completes. $\square$

14 The Standard Library

Each `cfc` verb binds to a named semantic operation of Section 2–8. We state the contract (inputs, output, invoked operation); the operation’s construction is fixed by the semantic model.

Definition 14.1 (ADMINISTER). `measure p yield r` invokes `ADMINISTER( $\Sigma, p$ )`: integrate each probe of panel p against the forward model from the current state, returning the reading r (the species distribution at read time, Theorem 2.2). The reading is a `Reading` value carrying its floor annotation.

Definition 14.2 (SURVIVE, LOCALIZE). `localize x using p until resolved yield c` invokes `SURVIVE( $\Sigma, p$ )`: accumulate the panel’s readings into the surviving set $\mathcal{M}_S$ (Theorem 6.5) and admit when cell-tight, yielding the `Cell` c ; on stable non-cell-tight $\mathcal{M}_S$ it declines (Theorem 13.8). The cell carries its coherence η and disease vector $\mathbf{D}$ (Theorem 5.6).

Definition 14.3 (GAP, SOLVE, CLOSE). `close gap of x toward coherent using basis b until coherent yield t` invokes `GAP( $x$ )` to compute the displacement to the nearest coherent state (Theorem 7.1), then `SOLVE` to solve the minimal- ℓ_1 program (Theorem 7.4) over basis b , applies the chosen doses, and iterates as a contraction (Theorem 7.7) until the gap is below the floor, yielding the `Intervention` t (the sparse dose vector with its predicted displacement). Infeasibility declines (Theorem 7.5).

Definition 14.4 (Soundness of the runtime). A `cfc` runtime is *sound* if (i) ADMINISTER agrees with the true reading up to the uniform derived-model bound of Theorem 9.2; (ii) SURVIVE returns a superset of the true surviving set (no false exclusion of the actual state); (iii) SOLVE returns a feasible intervention whose simulated displacement matches its physical displacement up to the same bound. Soundness reduces to the derived-model guarantee (Theorem 9.3).

Remark 14.5 (The verbs are one primitive). ADMINISTER, SURVIVE, and SOLVE are the single characterised-perturbation primitive read three ways: as a raw reading, as an accumulation of readings into a cell, and as a displacement chosen to close a gap. By Theorem 8.1 the measuring and moving faces are the same act; the standard library is therefore not three independent operations but one act exposed at three roles. No theorem depends on this reading.

15 Worked Programs

15.1 Localize a patient to a cell

Listing 1: Diagnosis is localization to a cell, never a label.

```
-- localize.cfc
floor 1e-3 -- ambient floor; nothing resolves finer

basis Metabolic {
  probe glucose_challenge {
    reference: healthy_ref
    observe: ingest("glucose", 75) >> glycolysis >> tca >> etc
  }
  probe lactate_probe {
    reference: healthy_ref
    observe: ingest("pyruvate", 10) >> ldh_branch
  }
}

patient P {
  measure panel { glucose_challenge() || lactate_probe() } yield r
```

```

-- run the rigidity loop: keep probing until one cell survives,
-- otherwise honestly decline (the evidence does not single out a state)
localize r
  using panel { glucose_challenge() >> tca_followup() }
  until resolved otherwise decline
  yield here

-- 'here : Cell' carries coherence and the disease vector; no disease name
assert here.coherence >= 0.0#1e-3
yield here
}

```

15.2 Close the gap without naming a disease

Listing 2: Therapy is gap-closure to the coherent set.

```

-- treat.cfc
floor 1e-3
import Metabolic

patient Q {
  localize baseline
  using panel { glucose_challenge() >> tca_followup() }
  until resolved
  yield here -- here : Cell (must be a Cell before 'close')

  -- close the measured gap toward coherence with the sparsest set of
  -- characterised interventions, chosen by effect, not by indication
  close gap of here
  toward coherent
  using basis Metabolic
  until coherent otherwise decline
  yield plan -- plan : Intervention (sparse dose vector)

  emit "gap closed to floor; no disease named"
  yield plan
}

```

15.3 Monitor continuously

Listing 3: A continuously working probe re-reads the state.

```

-- monitor.cfc
floor 1e-3
import Metabolic

patient R {
  monitor R every 3600 { -- re-run every 3600 time units
    measure panel { lactate_probe() } yield r
    localize r using panel { lactate_probe() } until resolved yield now
    -- raise concern iff coherence falls below threshold (Remark on concern)
    assert now.coherence >= 0.7#1e-3
    emit "cause for concern: coherence below threshold"
  }
}

```

Remark 15.1 (The same keyword set, three clinical jobs). Across the three programs, **measure** reads, **localize** diagnoses (to a cell, or declines), and **close** treats (to the floor, or declines). No program names a disease; each returns a cell, an intervention, or an honest decline. A reader who understands the characterised perturbation understands the whole language.

16 Complexity

Proposition 16.1 (Per-judgment complexity). *Let $n = \dim \mathcal{M}$, $m = |\mathcal{B}|$, $d = |\Omega|$, and let a cascade have depth ℓ and a reading return q species. Then:*

- (i) ***measure** costs one numerical integration of an n -dimensional system to the read time, plus $\mathcal{O}(q)$ to read back the species;*
- (ii) ***localize** costs $\mathcal{O}(q)$ constraint intersections against precomputed enabling sets per probe, and $\mathcal{O}(d)$ to compute the cell's coherence and disease vector (Theorem 4.5);*
- (iii) ***close** is a projection onto $\mathcal{M}^*$ plus an ℓ_1 -regularised convex program in m variables and n constraints, solvable in polynomial time (Boyd and Vandenberghe, 2004);*
- (iv) *the **close** contraction loop multiplies (iii) by $\mathcal{O}(\log_{1/\lambda}(1/\beta))$ steps (Theorem 7.7).*

Proof. (i) is one integration and a linear read-back. (ii) each probe adds one enabling-set intersection over q species; coherence and disease vector are $\mathcal{O}(d)$ by Theorem 4.5. (iii) the gap projection is a nearest-point computation onto the convex $\mathcal{M}^*$ and the dose solve is a convex program of the stated size. (iv) Theorem 7.7 bounds the number of contraction steps by $\lceil \log_{1/\lambda}(\text{dist}_0/\beta) \rceil$, each of cost (iii). $\square$

17 Constructive Validation

Because every object is a dynamical system on a finite-dimensional space and every quantity is an integration, a projection, an intersection, or a sparse program, each theorem is checkable by direct computation on explicit model instances, without clinical data. The following protocol validates the calculus and any conforming implementation (Sandve et al., 2013; Wilkinson et al., 2016).

1. **Floor** (Theorem 3.2). On a model Σ with fixed readout resolution δ , confirm the induced state-discrimination has a positive infimum β and that no state resolves below it.
2. **Coherence** (Theorem 4.4, Theorem 4.5). Confirm $\eta_i, \eta \in [0, 1]$ at optimal/degraded metric values and that the coherent set $\mathcal{M}_\theta^*$ is exactly the states with all $\eta_i \geq \theta$.
3. **Localization** (Theorem 5.4). Sample state pairs; confirm a floor-injective probe never maps distinct cells to the same reading within δ , and exhibit a non-injective probe that does, demonstrating insufficiency.
4. **Certificate** (Theorem 6.3). Construct a state-gated cascade; confirm species P_k appears iff every prior transition is state-enabled, and that states failing a transition produce P_k absent.
5. **Rigidity** (Theorem 6.5). Grow the observed species set and confirm $\mathcal{M}_S$ is non-increasing and becomes cell-tight exactly when no alternative cell survives.
6. **Gap-closure** (Theorem 7.4, Theorem 7.7). On states off $\mathcal{M}^*$, solve the sparse program and iterate the contraction; confirm geometric contraction of $\text{dist}(\cdot, \mathcal{M}^*)$ and sparsity of the selected intervention set.
7. **Unity** (Theorem 8.1). Confirm a single simulated perturbation yields both a localizing trace and a state displacement from one integration.

8. **Decline** (Theorem 13.8). Construct a model with two disjoint enabling clusters reachable by different probes; confirm `localize` reports *contested* decline when a probe separating them is withheld and *resolved* when it is supplied.
9. **Derived vs. fitted** (Theorem 9.2). Fit a surrogate model on a training region, then evaluate both surrogate and derived model off-region; confirm the surrogate’s trace error grows off-region while the derived model stays within its uniform bound—quantifying why free exploration requires the derived runtime.
10. **Type safety** (Theorems 12.4, 12.6 and 12.7). Confirm the type checker rejects (a) any literal or value of zero residue and (b) any `close` whose subject is not a `Cell`.

Each check is a finite, deterministic computation whose predicted and measured outcomes agree exactly for identities and within the stated bound for inequalities. The suite validates the calculus on explicit model instances; its fidelity to a real patient is inherited entirely from the fidelity of the derived forward model, the separate physical question the runtime must answer.

18 Reference Execution Model and Implementation

A conforming `cfc` implementation must provide, from this specification:

1. the shared front end: lexer (Section 10), parser to the grammar (Theorem 11.1), and accountability type checker (Section 12) enforcing Theorems 12.4 and 12.6;
2. the operational semantics of Section 13: a state-system configuration, a monotone committed-measurement count, and the `measure/localize/close/monitor` reductions, with evaluation mutating state (Theorem 13.3);
3. the standard-library bindings of Section 14 to `ADMINISTER`, `SURVIVE`, `SOLVE`;
4. a *derived* forward model (Theorem 9.1) satisfying runtime soundness (Theorem 14.4); a fitted surrogate may be offered only for exploration within its training region, and must be flagged as such.

Remark 18.1 (Reference interpreter). A direct interpreter of Section 13 over an exact solver—numerical integration for `ADMINISTER`, exact enabling-set intersection for `SURVIVE`, and a convex-program solver for `SOLVE`—is the simplest conforming implementation and serves as the oracle against which faster or approximate runtimes are checked. A conformance suite must verify the validation protocol of Section 17 and, in particular, that the type checker rejects zero-residue values and blind therapies.

19 Related Paradigms and Scope

`cfc` draws its formal machinery from three standard traditions and departs from clinical convention in one decisive respect.

From **programming-language theory** it inherits the discipline of a lexical grammar, a context-free grammar in EBNF, a static type system with progress-and-preservation soundness, and a small-step operational semantics (Aho et al., 2006; Pierce, 2002; Plotkin, 2004; Wright and Felleisen, 1994); its gradual-accountability flavour (every value carrying the floor at which it was determined) is a domain-specific refinement of the idea that types can track a quantitative invariant.

From **systems biology and constraint-based modelling** it inherits the view of physiology as a kinetic dynamical system and of intervention as constrained flux redirection (Kitano, 2002; Orth et al., 2010; Palsson, 2015); the demand that the runtime be derived rather than fitted

(Section 9) is the formal counterpart of the modelling community’s preference for mechanistic over regression models when extrapolation matters.

From **optimization** it inherits the sparse-selection geometry of the ℓ_1 program (Boyd and Vandenberghe, 2004; Candès et al., 2006; Tibshirani, 1996) and the contraction-mapping fixed-point theory (Banach, 1922) that make therapy a convergent, minimal-intervention procedure.

Its **departure from convention** is the elimination of the named disease. Where nosology partitions a continuous pathophysiology into catalogue entries (Feinstein, 1967; Loscalzo et al., 2007; Scheuermann et al., 2009), *cfc* returns a cell of state space and a measured gap. Naming, if desired, is a downstream description of which cell was occupied; it is never a prerequisite for either diagnosis or therapy (Theorems 3.5 and 7.8). The scope of the language is exactly this: to reduce clinical reasoning to localization and gap-closure over a derived state, relocating the clinician from precedent-retrieval to the specification of *intent*—which coherent target, how far, and at what cost to the particular person—the one input the calculus does not compute.

20 Conclusion

We have specified Cause-for-Concern as a language whose single primitive is the characterised perturbation—an administered compound of known composition read simultaneously for its fate (a measurement) and its displacement (an intervention)—and whose every operation follows from five proved facts. From a strictly positive resolution floor we obtained that a state is a cell, so diagnosis is localization to a cell rather than recognition against a disease catalogue (Theorem 3.2, Theorem 3.5). Health was defined intrinsically as membership of a coherent set, quantified by a scalar coherence and a disease vector, so no nosology is needed to say where health lies (Section 4). A probe localizes the state exactly when its reading map is floor-injective (Theorem 5.4); a cascading probe strengthens this to a trajectory certificate, since its later products exist only if the earlier path occurred, so a false account cannot forge the reading and the correct stopping rule is rigidity, not a threshold (Theorems 6.3 and 6.5). Therapy is the closure of the measured gap to the coherent set by the sparsest combination of characterised interventions selected by effect, executed as a contraction to a fixed point, with no disease ever named (Theorems 7.4 and 7.7). Diagnosis and therapy coincide, because every characterised perturbation both measures and moves, and the loop halts precisely when the state is coherent and rigidly known—the operative definition of wellness (Theorems 8.1 and 8.3).

On this foundation we built the language in full: a lexical and context-free grammar; an accountability type system in which no zero-residue value and no blind therapy are well-typed, proved sound by progress and preservation; a small-step operational semantics in which evaluation is measurement, with a monotone committed-measurement clock, proved deterministic and scheduler-sound; a standard library binding each verb to its semantic operation; a complexity account; a constructive validation protocol requiring no clinical data; and a reference execution model. The one substantive demand—that the forward model be derived, not fitted—was established as a theorem about extrapolation, because the safety of risk-free *in silico* experimentation reduces to it. The result is a deterministic, disease-free, report-free calculus of medicine in which the patient’s state is read from a self-certifying physiological program, the therapeutic target is a measured gap, and the clinician’s role is relocated from precedent-retrieval to judgment over a derived, explorable state. The specification is complete and implementable as written.

Data Availability

No empirical data was used in this specification. The validation protocol of Section 17 specifies how model instances and, subsequently, clinical data should be collected and compared to the language’s predictions.

Code Availability

Implementation requirements are given in Section 18. A reference interpreter realises the operational semantics of Section 13 directly over an exact solver.

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